



Real time face recognition system based on YOLO and InsightFace

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Abstract

Face Recognition is an important research topics in Machine Learning and Artificial Intelligence. It analyses and compares person's facial traits with a database that contains different faces to automatically recognise and verify a person's identity. It has attracted a lot of attention recently due to its non-intrusive nature. Unlike other biometric identification systems, face recognition does not require physical contact with the individual being identified, making it more convenient and hygienic. Although the existing face recognition system has achieved better performance, recognizing the obscured and disguised faces is difficult. Thus, to deal with these problems, this paper reveals a new real time unique face recognition network called YOLO-InsightFace that combines YOLO-V7, a cutting-edge deep learning model and InsightFace, one-of-a-kind 2D & 3D face analysis python module. YOLO-V7 is highly accurate and fast, making it ideal for real-time applications while InsightFace is capable of recognizing faces by generating highly discriminative face embeddings.

Keywords YOLO-V7 · InsightFace · Auto-Anchor Algorithm · Cosine Similarity Theorem · Arcface Loss · Eigenfaces

1 Introduction

Face recognition technology has emerged as a critical tool in security and surveillance applications, providing an efficient and reliable method of identifying individuals in real-time. It has made its way into various domains such as law enforcement, border control, access control, and authentication systems. The concept of face recognition is based on the fact that each individual has a unique facial structure that can be distinguished from others. The system utilizes digital image processing algorithms and artificial intelligence (AI) [7, 9] techniques to identify and verify the identity of an individual. Traditional face detection and face recognition systems [6, 27] include VGG16 architecture based face detection model with shallow layers produce light computing system, FPGA model for

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classifications [24], face detection system based on Support Vector Machine(SVM) techniques and Principle Component Analysis(PCA) [20] which are statistical methods used to reduce the number of variables in face identification. The modern face recognition tasks utilizes deep learning techniques, such as convolutional neural networks (CNNs) [16], that can learn to extract highly discriminative facial features from face images, enabling accurate face recognition even in complex scenarios.

The working principle of the face recognition system involves capturing an image or a video frame of an individual's face, which is then processed to extract the facial traits. These traits are then used to identify the person by comparison with database attributes of previously saved faces. The system may use various techniques such as Eigenfaces, local binary patterns (LBP) [18], and deep learning methods like convolutional neural networks (CNNs) to extract and match facial features. Face recognition system has several advantages over traditional identification methods such as identification cards, passwords, and PINs. One significant advantage is that it does not require physical contact, making it more hygienic, especially during pandemics. It also eliminates the possibility of fraud or theft of identification credentials, and helps in forensic investigation [15], making it a more secure and alternative technology. In law enforcement, face recognition technology can help identify and track suspects and criminals, as well as locating missing persons. In access control, face recognition technology can be used to grant access to restricted areas or secure buildings, enhancing the safety and security of the premises. In attendance tracking, the face recognition system can provide an accurate and efficient way of tracking attendance and ensuring that the right people are present at the right time.

In this paper, we reveal a latest iteration of face recognition system that uses two advanced algorithms for face detection and recognition. Firstly, the model uses YOLO-V7 algorithm for effective face detection and then followed by another attractive algorithm called InsightFace for facial feature extraction and recognition.

The following are the contribution made by the proposed model:

1. The model is able to find the faces in disguise, faces constrained by various lightening effects, and occluded faces.
2. The model detects and recognizes the face from real time environment by training the model using custom dataset.
3. YOLO-V7 algorithm can detect the faces quickly and InsightFace can do high level feature extraction for training and similarity measuring thereby increasing the efficiency of the face recognition system.

The rest of the paper briefly discusses on the existing works and algorithms used in this model along with the datasets and training. The model is evaluated using various performance analysis metrics. The result obtained proves that the proposed model is very efficient in the field of face recognition task and can be used in real time environment.

2 Traditional face recognition methods

Due to its potential to increase convenience and security in a variety of applications, including access control systems, surveillance, and identification systems, the Face Recognition technology has drawn a lot of attention in recent years. There are various methods and algorithms that have been developed to perform face recognition, each with its strengths and limitations.

One of the most commonly used face recognition methods is the LBP method, which is based on Principal Component Analysis (PCA) [2]. This method involves capturing the variations in the facial features of a set of training images and using this information to construct a set patterns, which represent the principal components of the data. The face image can then be projected into a dimension reduction technique called PCA, where recognition is carried out using a LBP. This method is effective in dealing with plastic surgery face recognition, but it requires a large amount of training data and can be sensitive to occlusions and pose variations.

Another popular method is the Local Binary Pattern (LBP) method [27], which is based on the texture analysis of facial features. This method involves extracting the LBP features of a face image by dividing the image into small regions and computing a binary code for each region based on the texture features. The binary codes are then concatenated to form a feature vector, which is used for recognition using a classification algorithm such as SVM. This method is robust to changes in illumination and facial expressions and requires less training data than the eigenface method.

The Deep Learning approaches [4, 28] has also gained popularity in recent years for face recognition. This method involves training a DNN using a large dataset of face images to learn discriminative features that are robust to variations in lighting, pose, and expressions. The trained network can then be used to extract features from new face images and perform recognition using a classification algorithm. The advantage of this method is that it can learn highly discriminative features and can handle complex variations in facial features. However, it requires a large amount of training data and is computationally expensive.

Another approach is the 3D face recognition method, which involves capturing the three-dimensional shape of the face using specialized sensors such as 3D cameras or structured light scanners. The 3D face shape can then be used for recognition, either by directly comparing the 3D shape or by extracting features such as depth maps or curvature information. This method is robust to changes in lighting and texture and can handle pose variations. However, it requires specialized hardware and can be sensitive to occlusions. Hybrid models using HOG-SVM networks are also used for face detection and recognition [17].

In addition to these methods, there are also other approaches like deep fake FR Systems [30] GAN [7], RetinaNet [11], SOM neural networks [13] that uses libraries like OpenCV [14] for computer vision task and combines multiple methods to achieve better performance. For example, the fusion of 2D and 3D face recognition methods can improve recognition performance by leveraging the strengths of each method.

YOLO (You Only Look Once) is a deep learning algorithm that can be used for face recognition. It works by creating a grid from the input picture and predicting bounding boxes and class probabilities for each grid cell. It can recognize multiple faces in an image and can perform real-time face recognition. YOLO face recognition is efficient and accurate, but it requires a large amount of training data and is computationally expensive. It is employed in various areas, including access control, surveillance, and picture retrieval. The latest FR system used YOLO-V5 [12] for face detection [1]. Different versions of YOLO such as YOLO-V3 [31] and YOLO-V4 [10] has made a various improvement in face detection technology. Overall, YOLO is a powerful tool for face recognition that can be used in a wide range of applications.

However, Face recognition is a complex problem that requires robust and accurate methods to achieve reliable results. The strategy chosen will rely on the particular application and the resources available. Research on existing FR Systems [29] has proved that every existing method has its strengths and limitations, and the performance can vary depending on the quality and quantity of the training data. Therefore, careful consideration and evaluation are necessary when selecting a face recognition method for a particular application.

3 Related works

A broad variety of face detection techniques are enhanced from object detection algorithms. The traditional face identification algorithms use handcrafted features to fulfil detection tasks before deep learning was introduced to this discipline. Many researchers provided a variety of detection methods to make up for feature representations limitations. However, complex designs are constantly required to speed up the detection process. The expression capability and computing efficiency of facial features have a significant impact on the behaviour of the detectors. The representative face detectors based on these handcrafted traits, such as Viola-Jones, Deformable Part Model (DPM) and Histogram of Oriented Gradients (HOG) are typical techniques.

Aria Satia Dharma et.al [18] introduced a useful method to reduce the complexity of the faces in face recognition that uses canny edge detection method and the LBP feature extraction method for detecting the faces and Self Organizing Map(SOM) to classify the faces.

Zainuddin et.al [19] implemented a face recognition system using Viola-Jones, Local Binary Pattern Histogram, and Euclidean Distance. The Viola-Jones method is used to detect the face image, the Local Binary Pattern Histogram is used to extract the features of the face, and Euclidean Distance is used to recognize the face. The model works in real time with good accuracy rate and fast detection time.

Giji George et.al [20] work focused on recognizing the faces before and after plastic surgery using Principal Component Analysis (PCA). PCA employs the Eigenface technique to address the 2-D face picture identification issue. The recognition rate of the system is least affected by expression and pose of input image.

Wanghua Deng et.al [5] proposed a face recognition system that uses MC-KCF algorithm to track the driver's face and recognize the facial key regions based on key-point detection. The Histogram Equalisation (HE) algorithm and illumination enhancement approach are employed to preprocess the picture frame's brightness, which increases the detection accuracy.

Wenqi Wu et.al [3] revealed a Different Scales Face Detector based on Faster R-CNN using a multitask RPN that extracts the high quality face proposals and coupled with parallel type Faster R-CNN which has three networks that works based on the proposal scales. Experimental results shows that this model has achieved an AP of 98% and recall rate of 93% on Large Scale Faces and works on real time.

K G Shanthi et.al [21] develops a face recognition model that aids the task force to track the missing people and for surveillance using Haar Cascade. It is a face detection method that is used to recognise faces in real-time videos. The Local Binary Pattern Histogram (LBPH) technique is used to identify faces. The drone produced a 98 percent accuracy rate when identifying people, which has important applications.

Suleman Khan et.al [22] described the advantage of using Transfer Learning in face recognition. The model is able to classify 1000 different people accurately comparing to other techniques like Fisherfaces, Eigenfaces, and LBPH.

To enhance the performance for face detection, Weijun Chen et al. [8] suggested a face detector called YOLO-face based on YOLOv3. The upgraded detector maintained its quick detection speed while greatly increasing accuracy. The detector, which is regarded as the best detector to date, improves performance by more than 50% when compared to other conventional algorithms.

Suman Menon M et al. [31] investigated several YOLO algorithm iterations, and an existing face detection and identification algorithm (R-CNN), and draw conclusions based on the performance of the YOLO-V3 algorithm for identifying a specific face.

Brandon Amos et al. [23] demonstrated the LFW benchmark performance of OpenFace and introduced a new classification benchmark for mobile settings. The document offers a brief introduction to the deep neural network methods and is aimed at non-experts interested in using OpenFace.

E Sugiharti, et al. [26] analyze the Facial Image Recognition using Two-Dimensional Principal Component Analysis which is coupled with KNN. In order to achieve a good accuracy value for the face recognition problem, the model employs the Two-Dimensional Principal Component Analysis (2DPCA) approach for feature extraction and the KNN classification method for data classification.

Bahjat Fatima et al. [25] developed two algorithms that enable accurate face and eye detections even when the driver is not facing the camera or the driver's eyes are closed. Additionally, they proposed a low-cost method of detecting driver drowsiness based on micro-sleep patterns. Using SVM and Adaboost, the classification is performed on the right eye exclusively to ascertain if the eye is closed or open. Micro-sleep patterns are identified based on eye conditions, and when necessary, an alarm is set off to alert the driver.

4 Proposed Model

Though the traditional algorithms have higher performance rate, the face recognition's efficiency is constrained by poor image quality. Also, it is difficult for the present models to accurately extract facial traits. Hence to address these problems, we therefore suggest a face recognition system that integrates the YOLO V7 framework with InsightFace. Real time face detection advances with new release of YOLO-V7 which is the latest version of YOLO. Insightface on the other hand, helps on high level feature extraction and similarity checking. The new model improves the efficiency of the face recognition system by identifying and extracting information even from obscured face images and thereby increases the recognition accuracy. As discussed earlier, the model uses YOLO-V7 algorithm for face detection task and InsightFace for training and recognition. InsightFace uses Resnet-100 for high level feature extraction and Cosine similarity theorem for feature similarity checking. The dataset used here is a custom dataset that contains 12 different classes with more than 2500 images. Figure 1 shows the block diagram representation of proposed model.

4.1 You Only Look Once (YOLO)

YOLO is a fully connected neural network-based single-stage object detector. Backbone, Head, and Neck make up the overall framework of YOLO. The Backbone primarily extracts the most crucial details from a picture and sends them to the Head through the Neck. The Neck creates feature pyramids by compiling feature maps that the Backbone has retrieved. The head also includes output layers with final detections. In the proposed model session, we go over how to train and recognise faces using YOLO-V7, the most recent iteration of the YOLO family.

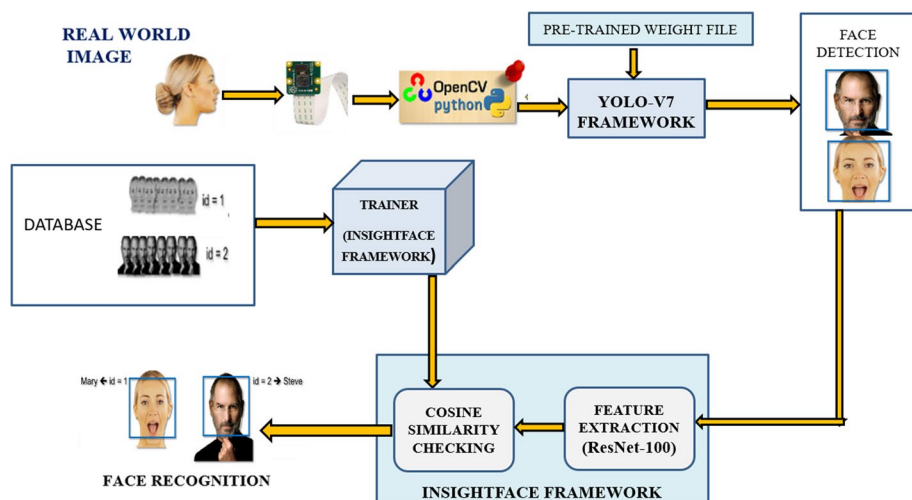


Fig. 1 Block diagram representation of the proposed face recognition model.

4.2 YOLO-V7 framework

YOLO-V7 is the latest version of YOLO family. This version improves speed and accuracy of object detectors by introducing several architectural reforms such as E-ELAN (Extended Efficient Layer Aggregation Network) Model Scaling for Concatenation-based Models. E-ELAN is responsible for computation and it is designed by considering the factors such as Memory access cost, I/O channel ratio, Element wise operation, Activations, Gradient path that affects the speed and accuracy. Figure 2 shows the architectural reforms of E-ELAN. It makes use of the cardinality expand, shuffle, and merge operations to enhance model learning without obliterating gradient flow routes. Regarding the architecture, it solely makes changes to the computational block while maintaining ELAN's design in the transition layer. The channels and cardinality of the computing block are increased via group convolution in E-ELAN. All of the computational blocks in a computing layer have the same channel multiplier and group parameter applied to them. Each computational block's feature map will be concatenated after being divided into groups of size g . To conduct merge cardinality, a shuffled group feature map will then be included. Compound Model Scaling in YOLO-V7 is used to make the model fit in various computing devices by considering the parameters such as resolution, width, depth and stage. This model also uses Re-parameterization after training to increase the inference by increasing the training time. Finally, the model uses Coarse-to-Fine Lead Head Guided Label Assigner and Lead Head Guided Label Assigner to that increases the learning capability.

4.2.1 Focal loss function

A brand-new loss function termed focal loss is used by YOLO-V7. To have a deep look on the model's learning on difficult misclassified samples, this modified form of cross entropy loss is employed. Older iterations of YOLO relied on a cross-entropy loss algorithm that wasn't as good at spotting little faces. Focal loss resolves this problem by

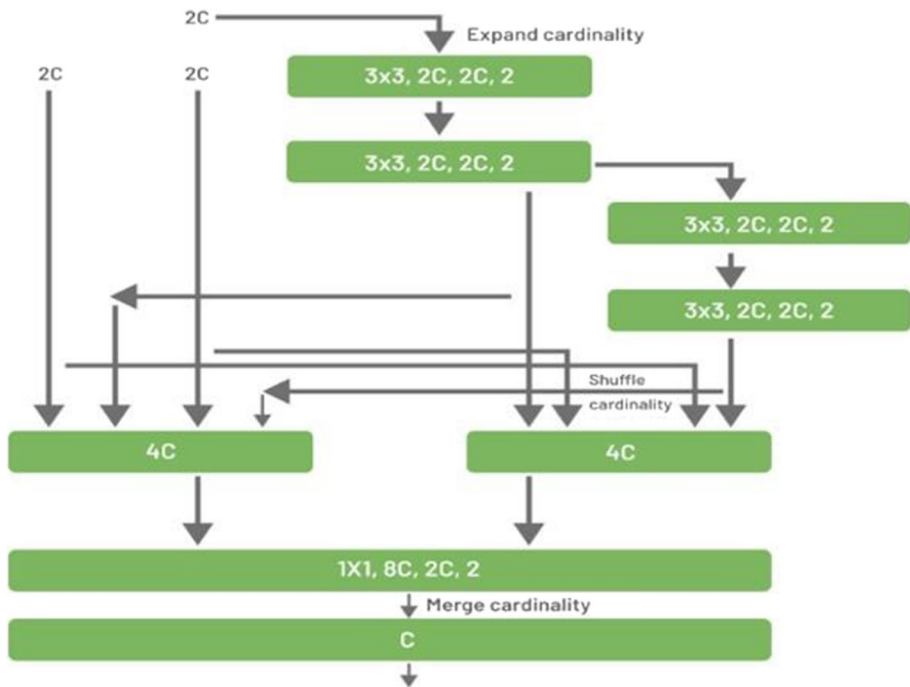


Fig. 2 Architecture of Extended Efficient Layer Network

down-weighting the loss for samples that have been correctly identified and concentrating on the difficult-to-detect hard cases. YoloV7 analyses photos at a resolution of 608 by 608 pixels, allowing it to find increasingly tiny faces. The focal loss function is given as:

$$FL(p_t) = -(1 - p_t)^\gamma \log p_t$$

Where p_t is the predicted probability from the model and γ is the tuneable focusing parameter. The simple well-classified instances (background) and the complex misclassified examples (foreground) contribute differently to the total loss function when is set to a positive integer. As a result, the optimisation method is forced to focus more on the challenging situations.

4.2.2 Auto-anchor algorithm

YOLO-V7 uses Auto-anchor algorithm. The idea behind the auto anchor algorithm is, before training the script checks how well provided anchors fit the data, and if they do not fit well, the script recalculates them; and the model is trained with new, more appropriate anchors. Initially, K-means is used to get the initial guess for anchor boxes. YOLO-V7 evolution runs for 1000 iterations (by default) and may change the initial anchor set by a lot. Later, an evolutionary algorithm is used to find the best anchor set with the respect to metric selected previously.

4.2.3 Intersection Over Union (IoU)

The overlap area to the total area of ground truth and prediction is measured by the IoU. IoU assesses the overlap of the Ground Truth and Prediction area in the context of face detection. IoU values are between 0 and 1, where 1 denotes complete overlap and 0 denotes no overlap at all. Thus, we can determine if a prediction is True Positive (TP), False Positive (FP), or False Negative (FN) with the use of the IoU threshold. General representation of IoU is given in the form,

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

4.2.4 Non-maximum suppression

NMS is a method mostly employed in the detection of objects that seeks to choose the best bounding box from a collection of overlapping boxes. NMS is a post-processing method that improves the accuracy and efficacy of face detection. During face detection, several bounding boxes are commonly created for a single item in an image. Even if they overlap or are located in different locations, these bounding boxes all represent the same item. NMS is used to locate and eliminate any extraneous or inaccurate bounding boxes in order to generate a single bounding box for each face in the picture.

4.3 InsightFace:

InsightFace is a DL-based algorithm for face recognition and analysis. The main idea behind the InsightFace algorithm is to learn discriminative features for face recognition using a large-scale dataset of faces. The algorithm uses a deep neural network architecture, specifically a variant of the ResNet architecture called ResNet-100, which has been pretrained on a large dataset of faces and high-level characteristics from the photos are extracted. Further, The main feature called ArcFace loss, is used by InsightFace. It pushes the neural network to acquire characteristics that are more discriminative for facial recognition. The ArcFace loss is designed to maximize the cosine similarity between features from the same identity while minimizing the cosine similarity between features from different identities. In addition to face recognition, InsightFace can also be used for face detection, face alignment, and face attribute analysis. InsightFace has a wide range of potential applications, including security and surveillance, biometrics, and social media. Its ability to recognize faces with high accuracy even in challenging conditions, such as low-resolution images, make it a promising technology for real-world applications.

4.3.1 Cosine similarity theorem

The cosine similarity theorem is a mathematical concept that measures the similarity between two vectors in a multi-dimensional space. It is commonly used in machine learning and natural language processing to compare the similarity between documents, images, or other data. In face recognition system, cosine similarity helps to check if

the two faces are identical or not. The cosine similarity between the two vectors can be computed as below:

$$\cos(\theta) = (\vec{A} * \vec{B}) / (\|\vec{A}\| * \|\vec{B}\|)$$

where A and B are the two vectors being compared. The value of the cosine similarity ranges from -1 to 1, where a value of 1 indicates that the two vectors are identical, a value of 0 indicates that the two vectors are orthogonal (i.e., they have no similarity), and a value of -1 indicates that the two vectors are diametrically opposed (i.e., they are completely dissimilar). It allows us to determine how similar two vectors are based on the angle between them, and it is an efficient way to compare large datasets (Fig. 3).

4.3.2 ArcFace loss

Additive Angular Margin Loss [32], often known as ArcFace, is a loss function utilised in facial recognition applications. Traditionally, the softmax is employed for the face recognition jobs. The softmax loss function does not explicitly optimise the feature embedding to enforce increased similarity for intraclass samples and diversity for inter-class samples and affects deep face recognition under significant intra-class appearance variations. The ArcFace loss transforms the logits $W_j^T x_i = \|W_j\| \|x_i\| \cos \theta_j$, where θ_j is the angle between the weight W_j and the feature x_i . By doing l_2 normalising, the individual weight $\|W_j\|$ is fixed. Normalisation fixes the embedding feature and changes the scale to s . Predictions are only dependent on the angle between the weight and the feature after the normalisation step on features and weights. Thus, a hypersphere with a radius of s is used to spread the learned embedding features. Finally, an additive angular margin penalty of m is placed between x_i and W_{yi} to simultaneously increase the intraclass compactness and interclass disparity. The recommended additive angular margin penalty is equivalent to the geodesic distance margin penalty in the normalised hypersphere and hence it is called as ArcFace (Fig. 4).

$$L_3 = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{yi}+m))}}{\sum_{j=1, j \neq yi}^n e^{s \cos \theta_j} e^{s(\cos(\theta_{yi}+m))}}$$

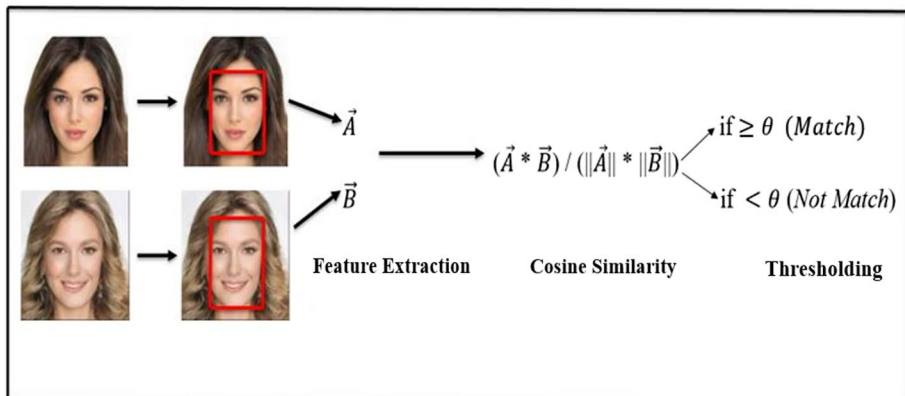


Fig. 3 Representation of feature extraction and feature matching verification

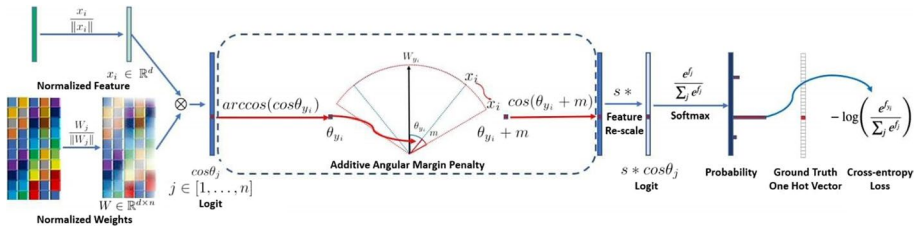


Fig. 4 [32] Training a DCNN for face recognition supervised by the ArcFace loss. Based on the feature x_i and weight W normalisation, we get the $\cos \theta_j$ (logit) for each class as $W_j^T x_i$. We calculate the $\arccos(\cos \theta_{y_i})$ and get the angle between the feature x_i and the ground truth weight W_{y_i} . In fact, W_j provides a kind of centre for each class. Then, we add an angular margin penalty m on the target (ground truth) angle θ_{y_i} . After that, we calculate $\cos(\theta_{y_i} + m)$ and multiply all logits by the feature scale s . The logits then go through the softmax function and contribute to the cross entropy loss.

5 Experiments

5.1 Dataset

Creating a custom dataset for face recognition is essential for building an accurate and effective system that meets specific requirements. In this work, we use our own dataset that consists of 12 classes. Each class consists of different person's face images with 300 images in each. The faces in the images need to be labelled manually by marking the coordinates of the face in each image. Various face patterns, including small scales, strong occlusions, and extreme postures are collected for each class in database. The images are uniformly resized and then trained using InsightFace algorithm. Figure 5 shows sample images of 3 classes in dataset.

5.2 Training

Training a model with InsightFace algorithm for face recognition involves several steps. Firstly, to train the model effectively, the dataset should include a diverse range of faces with variations in lighting, poses, and expressions. Secondly, the model architecture needs to be chosen and configured for face recognition. The InsightFace algorithm uses a DCNN with a specific loss function called the ArcFace loss. This helps in improving the accuracy of the model. Thirdly, the model should be trained on the dataset using techniques such as data augmentation and transfer learning. To improve the variety of the dataset, data augmentation entails adding random changes to the photos, such as rotations and flips. Transfer learning includes training a new model using a previously learned model as a starting point, which shortens training time and boosts performance. To evaluate the trained model's correctness and adjust any hyperparameters, it may then be tested on a different validation set (Fig. 6).



Fig. 5 Sample Images of Dataset

6 Results

To investigate YOLO-InsightFace model performance, we have done several studies. The model is evaluated on three different scenarios such as faces affected by various lightening conditions, faces under severe occlusion and disguised faces. The model is able to correctly recognize the faces on all the three different scenarios. The maximum rate at which the image processed is 55 fps in single GPU, whereas YOLO-V7 can also process upto 155 fps in multi GPU. Figure 7 shows the face recognition outputs of our proposed model on three different scenarios . Thus, the result shows that YOLO-InsightFace model outperforms the existing face recognition algorithms. Furthermore, the effect of our proposed model is evaluated using 4 object detection analysis metrics that are listed in Table 1,2,3 and 4.



Fig. 6 Results of proposed Face Recognition model on normal faces



Fig. 7 Results of proposed Face Recognition model on disguised and occluded faces.

7 Performance analysis of the proposed model

Table 1 Accuracy obtained for each class

Classes	Accuracy(%)
1	98.2
2	98.5
3	97.0
4	98.0
5	98.5
6	97.2
7	98.5
8	98.2
9	98.3
10	97.5
11	97.5
12	98.35

Table 2 Average Precision obtained for each class

Classes	Average Precision
1	0.985
2	0.985
3	0.985
4	0.985
5	0.984
6	0.984
7	0.984
8	0.983
9	0.982
10	0.980
11	0.975

Table 3 PR Curve for the proposed model

Recall	Precision
0.97	0.505
0.965	0.550
0.959	0.600
0.951	0.659
0.919	0.708
0.911	0.751
0.847	0.802
0.803	0.854
0.746	0.902
0.622	0.950
0.445	0.97

Table 4 Receiver Operating Characteristic (ROC)

Threshold	FPR	TPR
0.0	0.97	1
0.1	0.63	0.95
0.2	0.41	0.93
0.3	0.27	0.89
0.4	0.21	0.84
0.5	0.14	0.80
0.6	0.07	0.74
0.7	0.03	0.67
0.8	0.02	0.61
0.9	0.003	0.44
1.0	0	0.004



Fig. 8 Graphical Representation of mAP

8 Inferences

The above Figs. 8, 9, 10 shows the graphical representation of mAP, AUC and ROC Curve. However, the proposed model frequently produce predictions with a high level of confidence even for ambiguous or irrelevant inputs. Intuitively, both the estimation of uncertainty and the recognition of out-of-distribution (OOD) samples are advantageous to Face Recognition algorithms. If we look at the existing classification model from a probabilistic perspective, there are two advantages to YOLO-InsightFace face recognition model. (1) It can modify the learning strength during the training phase. (1) For accuracy and stability during training, the learning strength of clean and noisy samples can be changed. (2) Without training on additional labels, it can give a score of

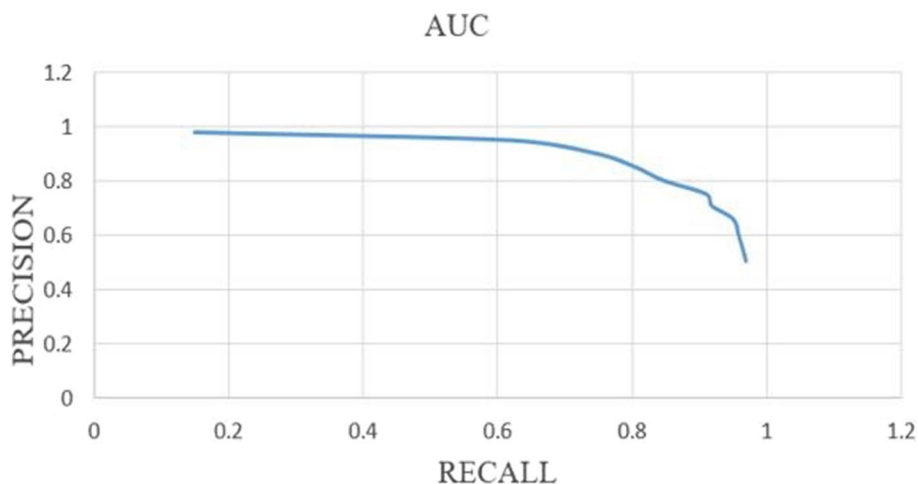


Fig. 9 AUC obtained from PR Curve

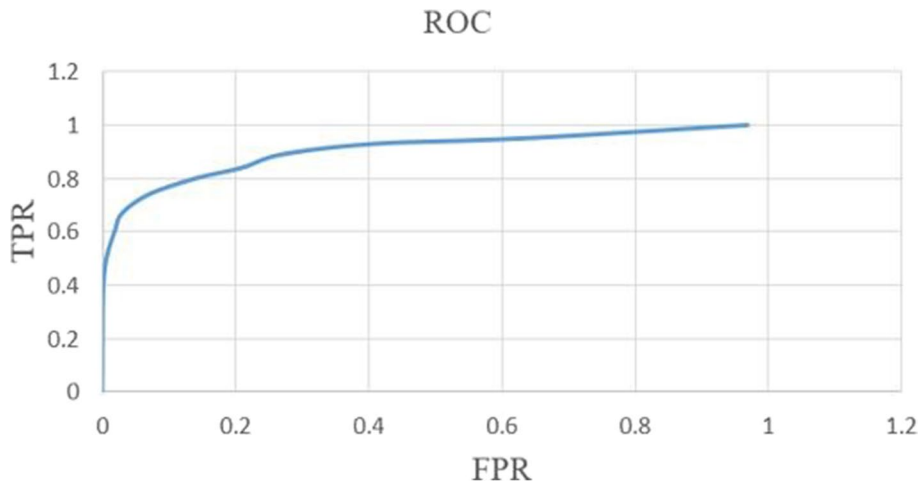


Fig. 10 ROC Curve for obtained TPR and FPR.

confidence to identify unsure and even OOD samples during the test phase. This model achieves an overall Accuracy of 98.5% which is far better than other existing face recognition models. The Precision Recall(PR) curve obtained has greater AUC of 90.2% comparing to the PR curves of other models. As the AUC is higher, the model performs perfectly by distinguishing between the positive and negative classes. Further, the Receiver Operating Characteristic(ROC) curve obtained for the respective True Positive Rate(TPR) and False Positive Rate(FPR) shows that the proposed model is more capable of classifying the images and the Mean Average Precision obtained from this work is 98.2% which is a another achievement of this model. This super small model can even recognize the small, disguised and occluded faces.

9 Comparison of YOLO-InsightFace with Existing Face Recognition techniques

It is clear from the above table (Table 5) that, the proposed YOLO-V7 and InsightFace FR system perform better in terms of speed and accuracy when compared to existing systems. The suggested system can even process pictures up to 155 frames per second using several GPU.

Table 5 Comparison of YOLO-InsightFace with existing FR techniques

Algorithms	Accuracy	FPS
Deepface	97%	50
Google Facenet	99.6%	15
SSD	79.8%	24
YOLO-V7 & InsightFace	98.5%	55 (single GPU)

10 Conclusions

In this work, we use YOLOv7 object detection algorithm that can accurately detect and localize faces in images and InsightFace, a deep learning algorithm that specializes in face recognition. It uses a specific loss function called the ArcFace loss, which maximizes the distance between different face embeddings while minimizing the distance between same-face embeddings. Model is trained on a custom dataset of face images with data augmentation and transfer learning techniques to improve accuracy and generalize well to unseen faces. Despite the high accuracy achieved by these algorithms, training and recognition speed is also quite better than other algorithms. The high accuracy and mAP achieved by face recognition using YOLOv7 and InsightFace algorithms demonstrate their potential for various applications. Both YOLO v7 and InsightFace, however, have advantages and disadvantages, and the best approach relies on the application at hand as well as the available resources. YOLO v7 is suitable for applications that require real-time face recognition, such as surveillance and access control systems. InsightFace is used for applications that needs high accuracy and robustness, such as identification systems. In recent years, there has been significant progress in the development of face recognition algorithms. The integration of deep learning with other techniques such as 3D face recognition and hybrid methods is expected to further improve the accuracy and efficiency of face recognition systems.

However, using face recognition technology also raises ethical and privacy concerns. Careful consideration must be given to the potential risks and implications of its use. To address these concerns, it is important for face recognition technology to be developed and deployed with transparency, accountability, and oversight. This includes ensuring that algorithms are trained on unbiased data, implementing appropriate privacy safeguards, and providing individuals with the ability to control their own personal data. It is a need to ensure that it is used responsibly and without bias. Thus, ongoing research and development are necessary to address these challenges.

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Data Availability Datasets analyzed during the current study is available from the corresponding author upon reasonable request.

Declarations

Consent for Publication The Author agrees to publish in the Journal indicated below and also to publication of the article in English by Springer in Springer's corresponding English-language journal.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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